Candy Mini Project

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Importing candy data

```r
candy_file <- "candy-data.csv"
candy = read.csv(candy_file, row.names= 1)
head(candy)
```

```
             chocolate fruity caramel peanutyalmondy nougat crispedricewafer
100 Grand            1      0       1              0      0                1
3 Musketeers         1      0       0              0      1                0
One dime             0      0       0              0      0                0
One quarter          0      0       0              0      0                0
Air Heads            0      1       0              0      0                0
Almond Joy           1      0       0              1      0                0
             hard bar pluribus sugarpercent pricepercent winpercent
100 Grand       0   1        0        0.732        0.860   66.97173
3 Musketeers    0   1        0        0.604        0.511   67.60294
One dime        0   0        0        0.011        0.116   32.26109
One quarter     0   0        0        0.011        0.511   46.11650
Air Heads       0   0        0        0.906        0.511   52.34146
Almond Joy      0   1        0        0.465        0.767   50.34755
```

What is in the dataset?

Q1: Variety

```r
dim(candy)
```

```
[1] 85 12
```

```r
nrow(candy)
```

```
[1] 85
```

There are a total of 85 different candies in this set.

Q2: Fruity Candy

```r
table(candy$fruity)
```

```

 0  1 
47 38 
```

```r
sum(candy$fruity)
```

```
[1] 38
```

Out of the 85 candies, 38 of them are fruity type.

What is your favorite candy?

Q3: My favorite candy

```r
candy["Snickers",]$winpercent
```

```
[1] 76.67378
```

Snickers has a favorable 76.67% compared to other candy.

Q4: Kit Kat

```r
candy["Kit Kat",]$winpercent
```

```
[1] 76.7686
```

Kit Kat ranks higher than Snickers at 76.77% but lower than Twix.

Q5: Tootsie Roll Snack Bar

```
candy["Tootsie Roll Snack Bars",]$winpercent
```

```
[1] 49.6535
```

Almost half favor tootsie rolls at 49.65%

skimr::skim() functions

```
library("skimr")
skim(candy)
```

Table 1: Data summary

Name	candy
Number of rows	85
Number of columns	12
Column type frequency:	
numeric	12
Group variables	None

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
chocolate	0	1	0.44	0.50	0.00	0.00	0.00	1.00	1.00	
fruity	0	1	0.45	0.50	0.00	0.00	0.00	1.00	1.00	
caramel	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
peanutyalmondy	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
nougat	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
crispedricewafer	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
hard	0	1	0.18	0.38	0.00	0.00	0.00	0.00	1.00	
bar	0	1	0.25	0.43	0.00	0.00	0.00	0.00	1.00	
pluribus	0	1	0.52	0.50	0.00	0.00	1.00	1.00	1.00	
sugarpercent	0	1	0.48	0.28	0.01	0.22	0.47	0.73	0.99	
pricepercent	0	1	0.47	0.29	0.01	0.26	0.47	0.65	0.98	
winpercent	0	1	50.32	14.71	22.45	39.14	47.83	59.86	84.18	

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist

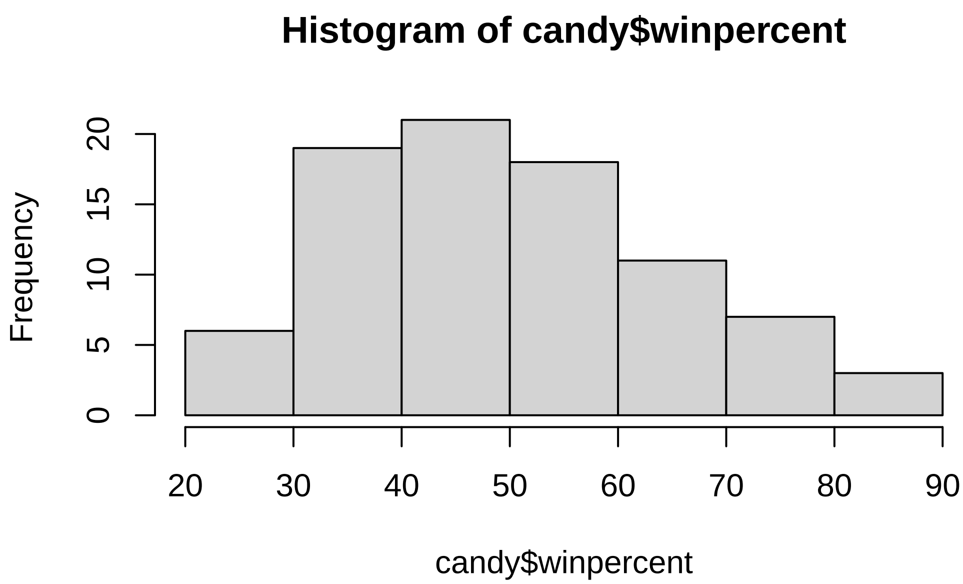
Q6: The Winpercent data is seemed to be distributed on a scale with averages rather than ranking in order of high to low.

Q7: 0 and 1's refer to Binary code that means true or false in our scenario. A 1 means it is a chocolate candy and a 0 is not.

Exploratory Analysis

Q8: Histogram and ggplot2

```r
hist(candy$winpercent)
```

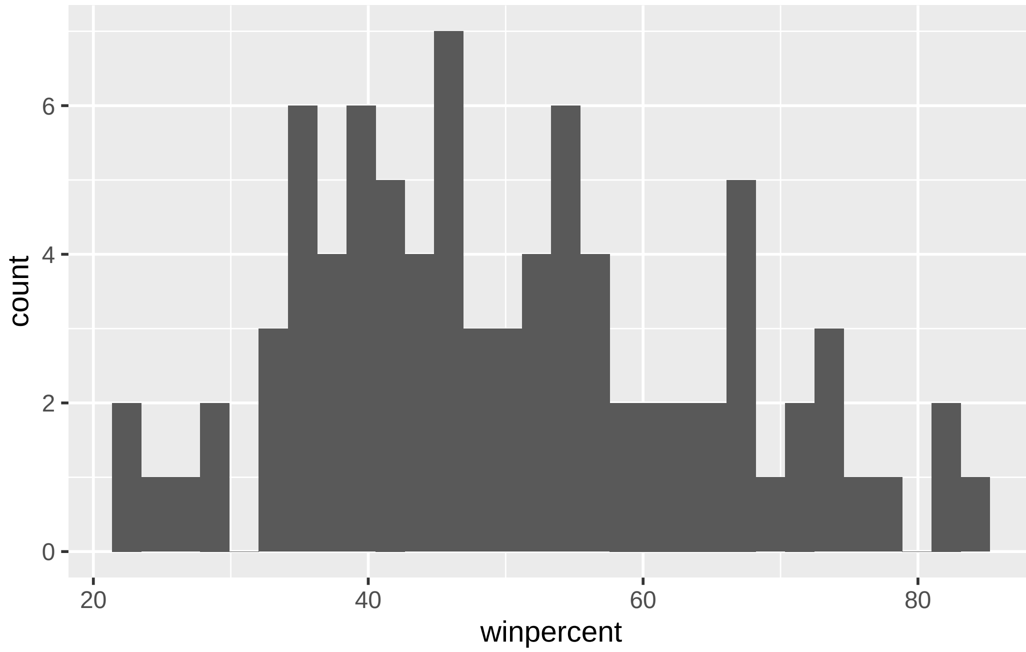



```r
library(ggplot2)

ggplot(candy, aes(x = winpercent)) +
  geom_histogram(bindwidth = 10)
```

```
Warning in geom_histogram(bindwidth = 10): Ignoring unknown parameters:
`bindwidth`
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



Q9: Are the values symmetrical?

Based on the data, the values are skewed to the right and not symmetrical towards the middle

Q10: Above or Below 50

The center is below 50% because there are higher value counts in the 40% range.

Q11: Chocolate Ranking

```r
mean(candy$winpercent[as.logical(candy$chocolate)])
```

```
[1] 60.92153
```

```r
mean(candy$winpercent[as.logical(candy$fruity)])
```

```
[1] 44.11974
```

Chocolate is ranked higher than Fruit candy with a 60.92% favored versus 44.12%.

Q12: Statistical Significance

```
t.test(candy$winpercent[as.logical(candy$chocolate)],
       candy$winpercent[as.logical(candy$fruity)])
```

```
    Welch Two Sample t-test

data:  candy$winpercent[as.logical(candy$chocolate)] and candy$winpercent[as.logical(candy$fr
t = 6.2582, df = 68.882, p-value = 2.871e-08
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 11.44563 22.15795
sample estimates:
mean of x mean of y
 60.92153  44.11974
```

Yes, there is a statistical significance based on the alternative hypothesis being true.

Overall Candy Rankings

Q13: Least liked candies

```
head(candy[order(candy$winpercent),], n=5)
```

```
                   chocolate fruity caramel peanutyalmondy nougat
Nik L Nip                  0      1       0              0      0
Boston Baked Beans         0      0       0              1      0
Chiclets                   0      1       0              0      0
Super Bubble               0      1       0              0      0
Jawbusters                 0      1       0              0      0
                   crispedricewafer hard bar pluribus sugarpercent pricepercent
Nik L Nip                         0    0   0        1        0.197        0.976
Boston Baked Beans                0    0   0        1        0.313        0.511
Chiclets                          0    0   0        1        0.046        0.325
Super Bubble                      0    0   0        0        0.162        0.116
Jawbusters                        0    1   0        1        0.093        0.511
                   winpercent
Nik L Nip            22.44534
```

```
Boston Baked Beans   23.41782
Chiclets             24.52499
Super Bubble         27.30386
Jawbusters           28.12744
```

```r
library(dplyr)
```

```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```r
candy |> arrange(winpercent) |> head(5)
```

```
                   chocolate fruity caramel peanutyalmondy nougat
Nik L Nip                  0      1       0              0      0
Boston Baked Beans         0      0       0              1      0
Chiclets                   0      1       0              0      0
Super Bubble               0      1       0              0      0
Jawbusters                 0      1       0              0      0
                   crispedricewafer hard bar pluribus sugarpercent pricepercent
Nik L Nip                         0    0   0        1        0.197        0.976
Boston Baked Beans                0    0   0        1        0.313        0.511
Chiclets                          0    0   0        1        0.046        0.325
Super Bubble                      0    0   0        0        0.162        0.116
Jawbusters                        0    1   0        1        0.093        0.511
                   winpercent
Nik L Nip            22.44534
Boston Baked Beans   23.41782
Chiclets             24.52499
Super Bubble         27.30386
Jawbusters           28.12744
```

The bottom 5 candies were Nik L Nip, Boston Baked Beans, Chiclets, Super Bubble, and Jawbusters from least to most liked.

Q14: All Time Favs

```
head(candy[order(candy$winpercent, decreasing=TRUE),], n=5)
```

```
                          chocolate fruity caramel peanutyalmondy nougat
Reese's Peanut Butter cup         1      0       0              1      0
Reese's Miniatures                1      0       0              1      0
Twix                              1      0       1              0      0
Kit Kat                           1      0       0              0      0
Snickers                          1      0       1              1      1
                          crispedricewafer hard bar pluribus sugarpercent
Reese's Peanut Butter cup                0    0   0        0        0.720
Reese's Miniatures                       0    0   0        0        0.034
Twix                                     1    0   1        0        0.546
Kit Kat                                  1    0   1        0        0.313
Snickers                                 0    0   1        0        0.546
                          pricepercent winpercent
Reese's Peanut Butter cup        0.651   84.18029
Reese's Miniatures               0.279   81.86626
Twix                             0.906   81.64291
Kit Kat                          0.511   76.76860
Snickers                         0.651   76.67378
```

```
candy |> arrange(desc(winpercent)) |> head(5)
```

```
                          chocolate fruity caramel peanutyalmondy nougat
Reese's Peanut Butter cup         1      0       0              1      0
Reese's Miniatures                1      0       0              1      0
Twix                              1      0       1              0      0
Kit Kat                           1      0       0              0      0
Snickers                          1      0       1              1      1
                          crispedricewafer hard bar pluribus sugarpercent
Reese's Peanut Butter cup                0    0   0        0        0.720
Reese's Miniatures                       0    0   0        0        0.034
Twix                                     1    0   1        0        0.546
Kit Kat                                  1    0   1        0        0.313
Snickers                                 0    0   1        0        0.546
                          pricepercent winpercent
Reese's Peanut Butter cup        0.651   84.18029
```

```
Reese's Miniatures              0.279   81.86626
Twix                            0.906   81.64291
Kit Kat                         0.511   76.76860
Snickers                        0.651   76.67378
```

```
candy |> arrange(desc(winpercent)) |> head(5)
```

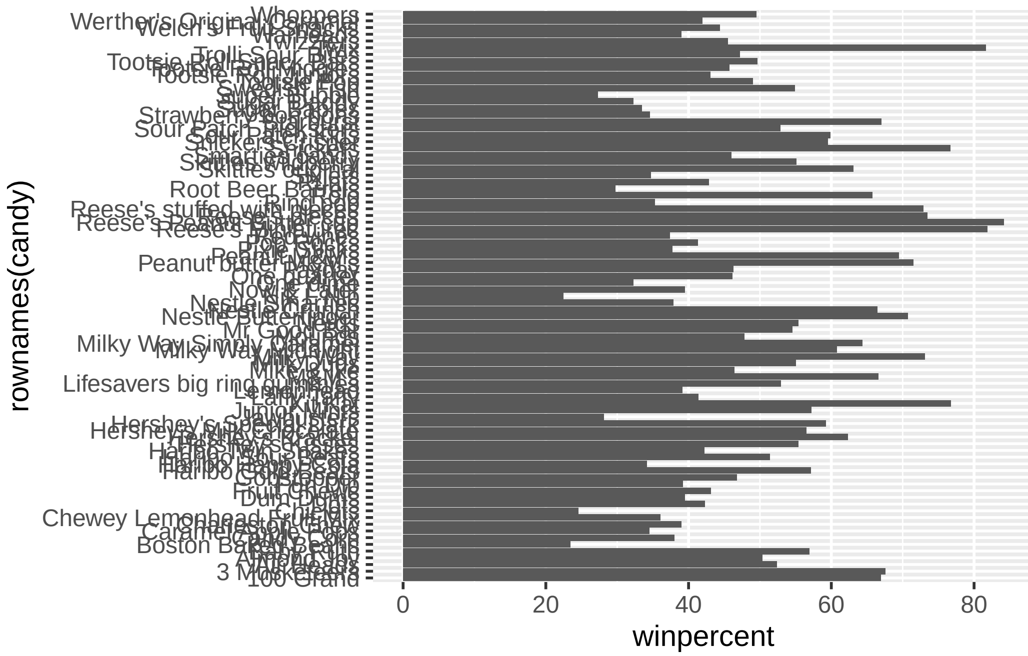
```
                          chocolate fruity caramel peanutyalmondy nougat
Reese's Peanut Butter cup         1      0       0              1      0
Reese's Miniatures                1      0       0              1      0
Twix                              1      0       1              0      0
Kit Kat                           1      0       0              0      0
Snickers                          1      0       1              1      1
                          crispedricewafer hard bar pluribus sugarpercent
Reese's Peanut Butter cup                0    0   0        0        0.720
Reese's Miniatures                       0    0   0        0        0.034
Twix                                     1    0   1        0        0.546
Kit Kat                                  1    0   1        0        0.313
Snickers                                 0    0   1        0        0.546
                          pricepercent winpercent
Reese's Peanut Butter cup        0.651   84.18029
Reese's Miniatures               0.279   81.86626
Twix                             0.906   81.64291
Kit Kat                          0.511   76.76860
Snickers                         0.651   76.67378
```

The top 5 favorites are Reese's Peanut Butter Cup, Reese's Miniature, Twix, Kit Kat, and Snickers from most to least.

The second option of dpylr seems it would be more preferable based on the direct compact line of code versus R's use of variables.

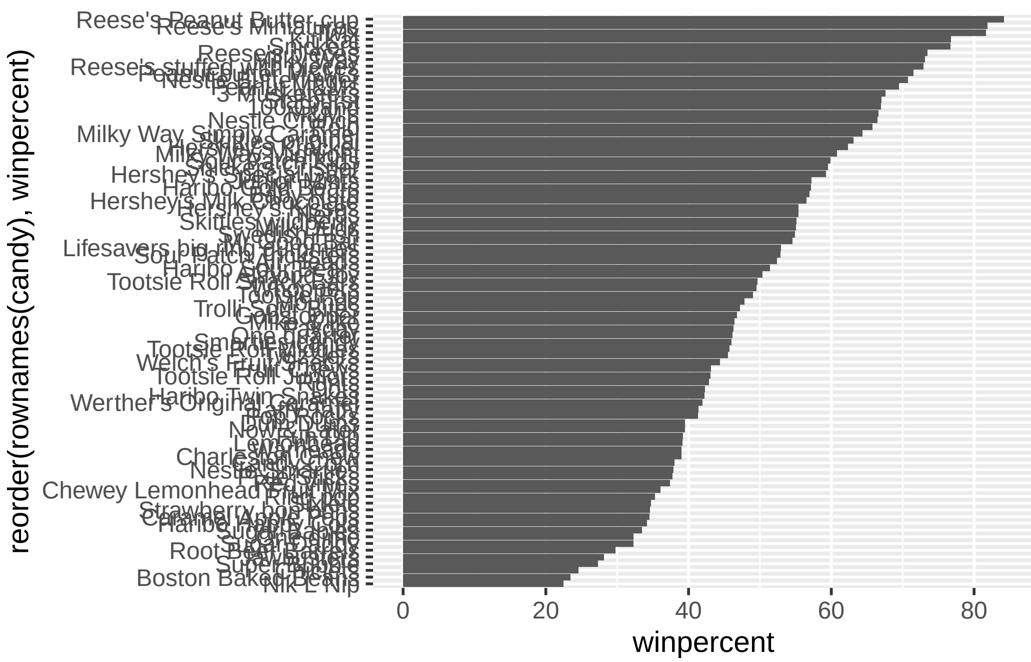
Q15: Barplot ranking

```
library(ggplot2)
ggplot(candy) +
  aes(winpercent, rownames(candy)) +
  geom_col()
```



Q16: Reordered

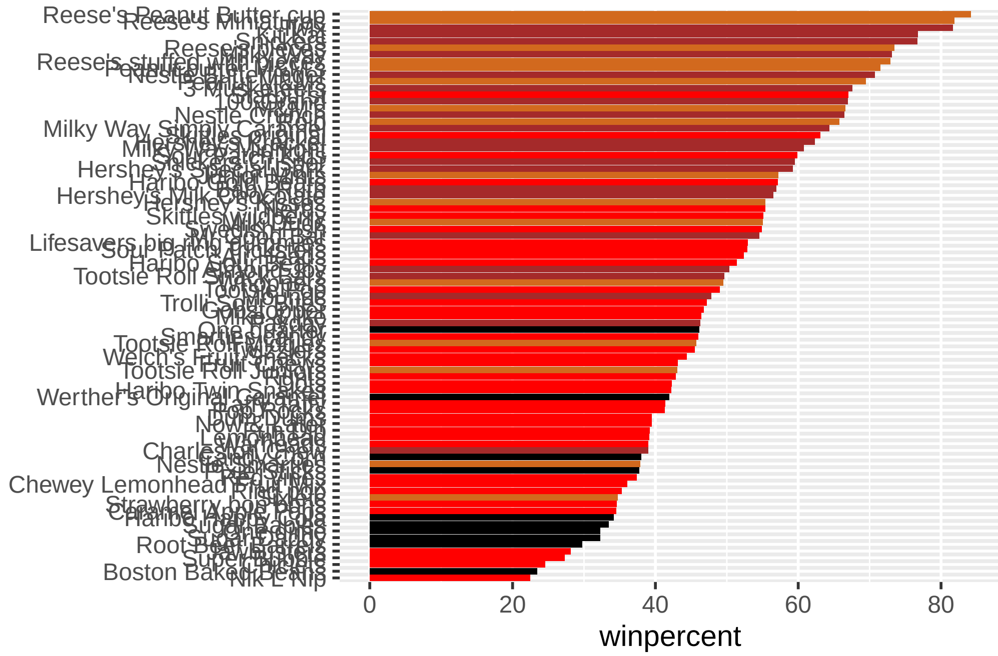
```
ggplot(candy) +
  aes(winpercent, reorder(rownames(candy), winpercent)) +
  geom_col()
```



Adding useful colors

```
my_cols = rep("black", nrow(candy))
my_cols[as.logical(candy$chocolate)] = "chocolate"
my_cols[as.logical(candy$bar)] = "brown"
my_cols[as.logical(candy$fruity)] = "red"
```

```
ggplot(candy) +
  aes(winpercent, reorder(rownames(candy), winpercent)) +
  geom_col(fill=my_cols) +
  ylab("")
```



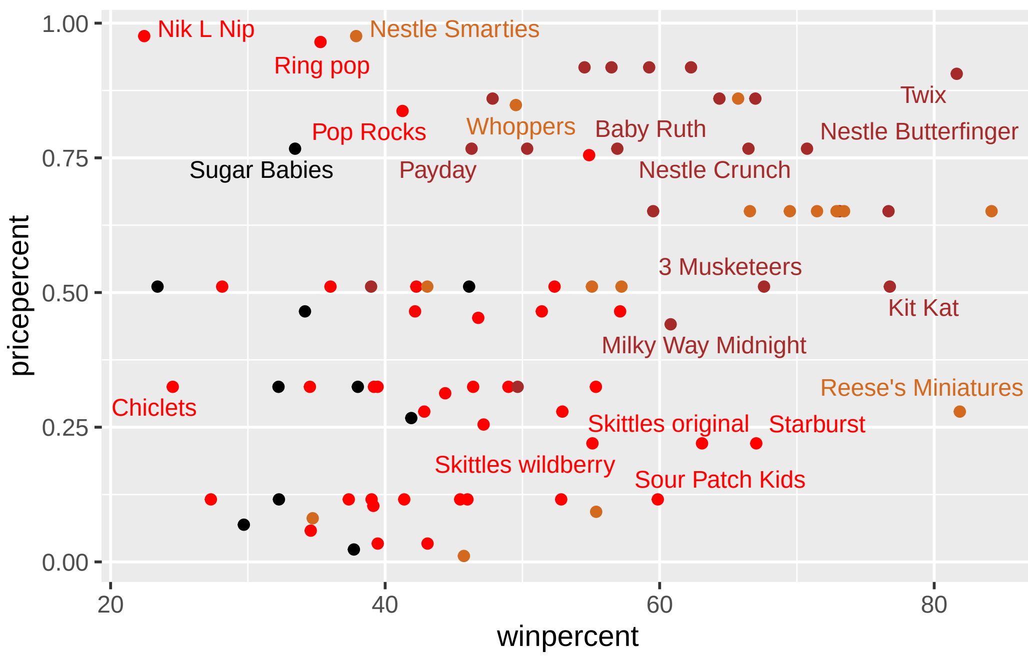
Q17: Sixlets is ranked the WORST chocolate candy

Q18: Starburst is ranked the BEST fruity candy.

Pricepoint

```
library(ggrepel)

# How about a plot of win vs price
ggplot(candy) +
  aes(winpercent, pricepercent, label=rownames(candy)) +
  geom_point(col=my_cols) +
  geom_text_repel(col=my_cols, size=3.3, max.overlaps = 5)
```



Q19: Best value

```r
ord <- order(candy$winpercent, decreasing = TRUE)
head( candy[ord,c(11,12)], n=5 )
```

```
                         pricepercent winpercent
Reese's Peanut Butter cup        0.651   84.18029
Reese's Miniatures               0.279   81.86626
Twix                             0.906   81.64291
Kit Kat                          0.511   76.76860
Snickers                         0.651   76.67378
```

Reese's Miniatures have the best value because they have a high winpercent with the lowest pricepercent overall.

Q20: Top 5 Most Expensive

```r
ord <- order(candy$pricepercent, decreasing = TRUE)
head( candy[ord,c(11,12)], n=5 )
```

```
                         pricepercent winpercent
Nik L Nip                        0.976   22.44534
```

```
Nestle Smarties                 0.976    37.88719
Ring pop                        0.965    35.29076
Hershey's Krackel               0.918    62.28448
Hershey's Milk Chocolate        0.918    56.49050
```

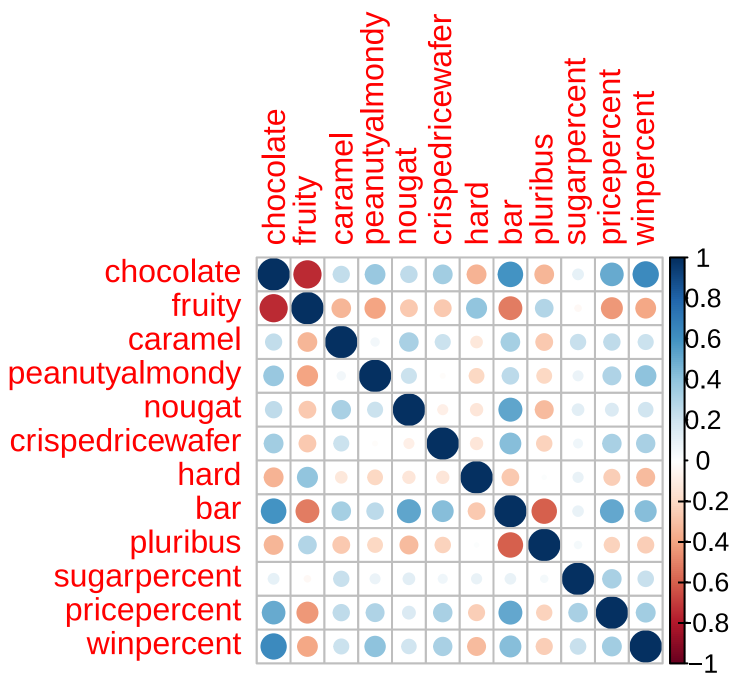
Nik L Nip would be the most expensive and least popular due to highest pricepercent (0.976) and lowest winpercent (22.45%) values. The other most expensive are Smarties, Ring Pop, Krackel, and Hershey's Milk Chocolate.

Correlation Structure

```r
library(corrplot)
```

```
corrplot 0.95 loaded
```

```r
cij <- cor(candy)
corrplot(cij)
```



Q22: Anti-Correlation

Chocolate and Fruity are two categories that are the least related with the big dark red circle in fruity (highly negative) and big dark blue circle in chocolate (highly positive). Since both are drastic, the candy options are either or but never both.

Q23: Largest Contributor to PC1

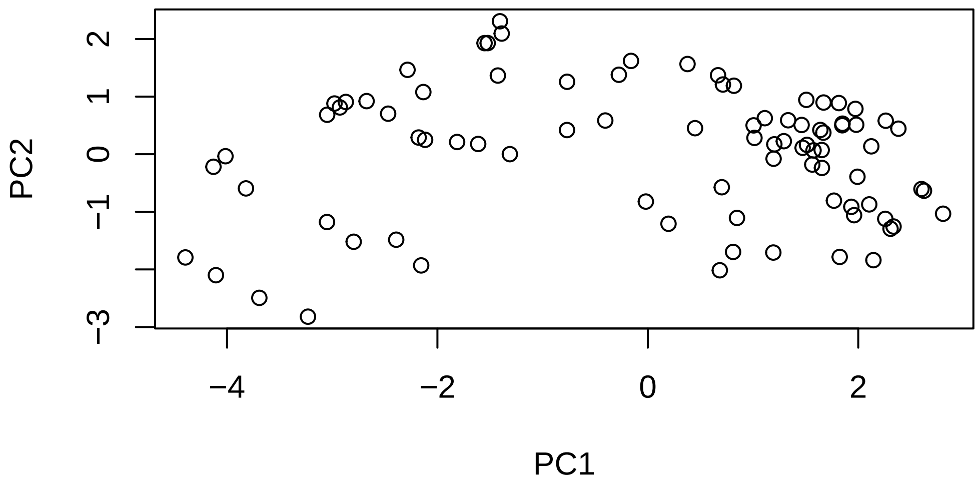
The competition of preference between Chocolate versus Fruit flavors in the data pool. I personally due like a combination like Blueberry Chocolate or Raspberry Chocolate, but I am one small statistic that may not contribute much.

Principal Component Analysis

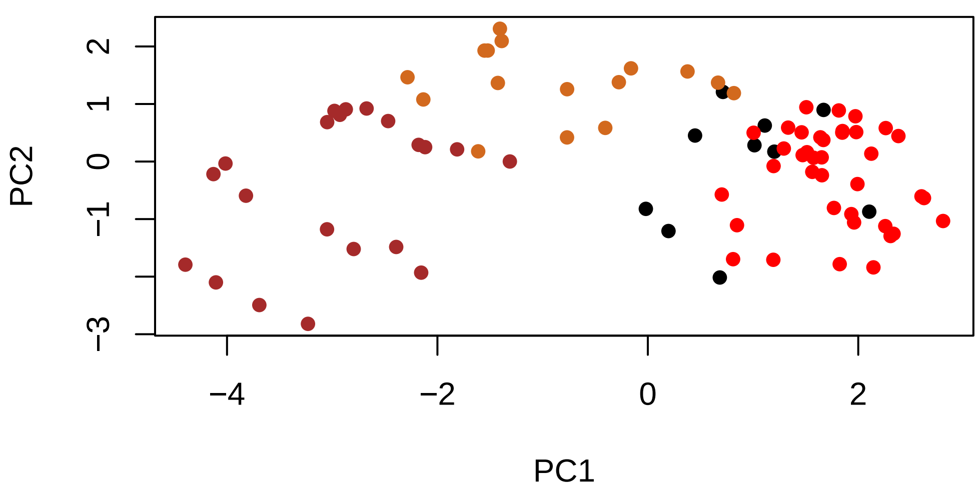
```r
pca <- prcomp(candy, scale=TRUE)
summary(pca)
```

```
Importance of components:
                          PC1    PC2    PC3     PC4    PC5     PC6     PC7
Standard deviation     2.0788 1.1378 1.1092 1.07533 0.9518 0.81923 0.81530
Proportion of Variance 0.3601 0.1079 0.1025 0.09636 0.0755 0.05593 0.05539
Cumulative Proportion  0.3601 0.4680 0.5705 0.66688 0.7424 0.79830 0.85369
                           PC8     PC9    PC10    PC11    PC12
Standard deviation     0.74530 0.67824 0.62349 0.43974 0.39760
Proportion of Variance 0.04629 0.03833 0.03239 0.01611 0.01317
Cumulative Proportion  0.89998 0.93832 0.97071 0.98683 1.00000
```

```r
plot(pca$x[,1:2])
```

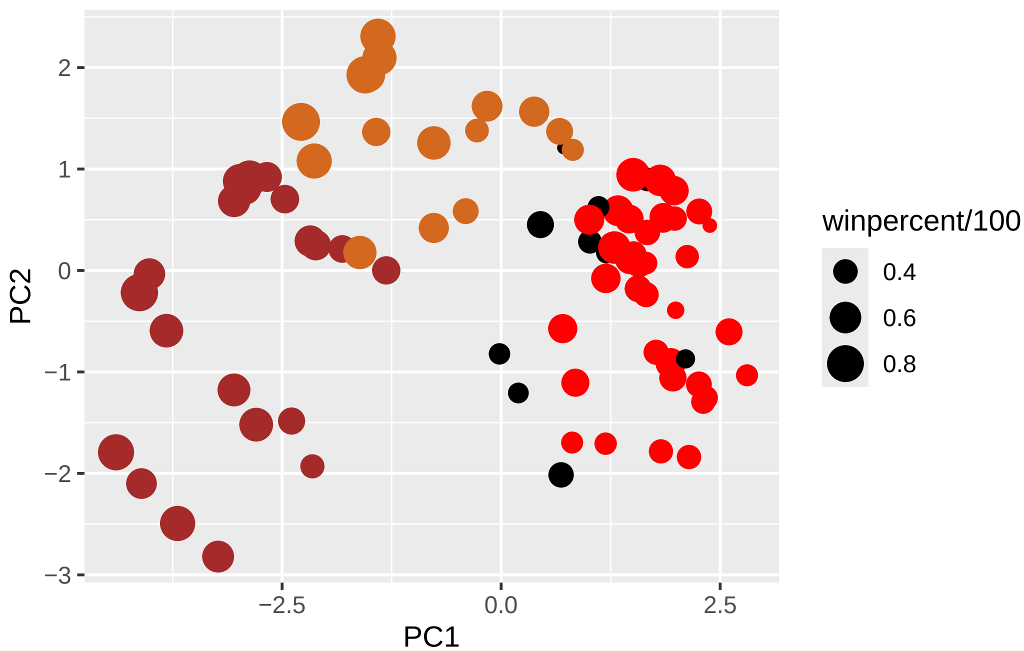



```
plot(pca$x[,1:2], col=my_cols, pch=16)
```



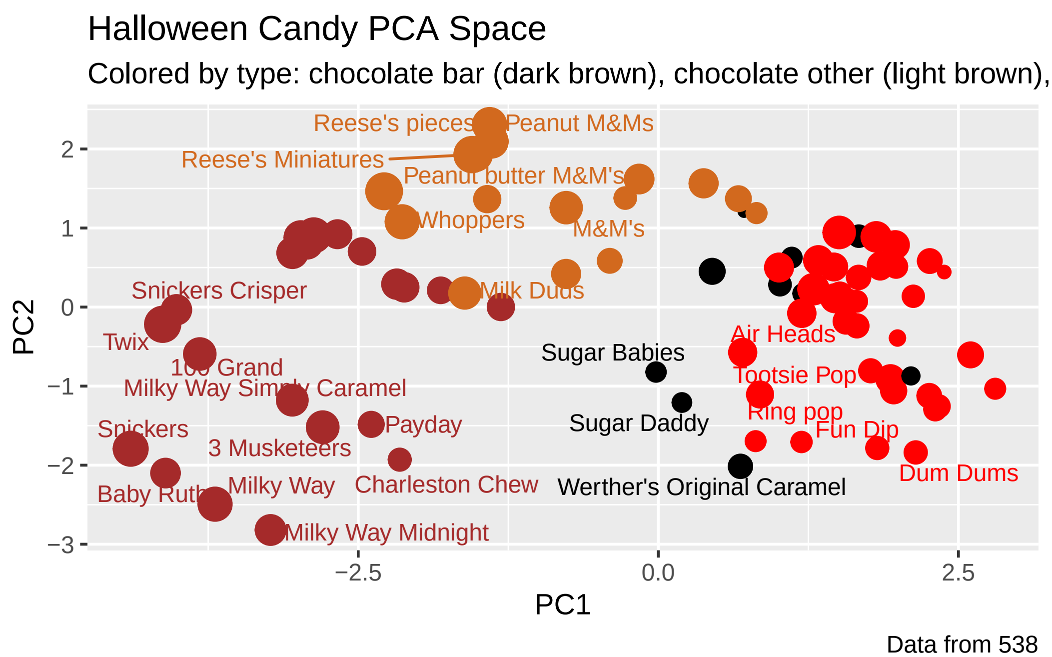
```r
# Make a new data-frame with our PCA results and candy data
my_data <- cbind(candy, pca$x[,1:3])
```

```r
p <- ggplot(my_data) +
        aes(x=PC1, y=PC2,
            size=winpercent/100,
            text=rownames(my_data),
            label=rownames(my_data)) +
        geom_point(col=my_cols)
p
```

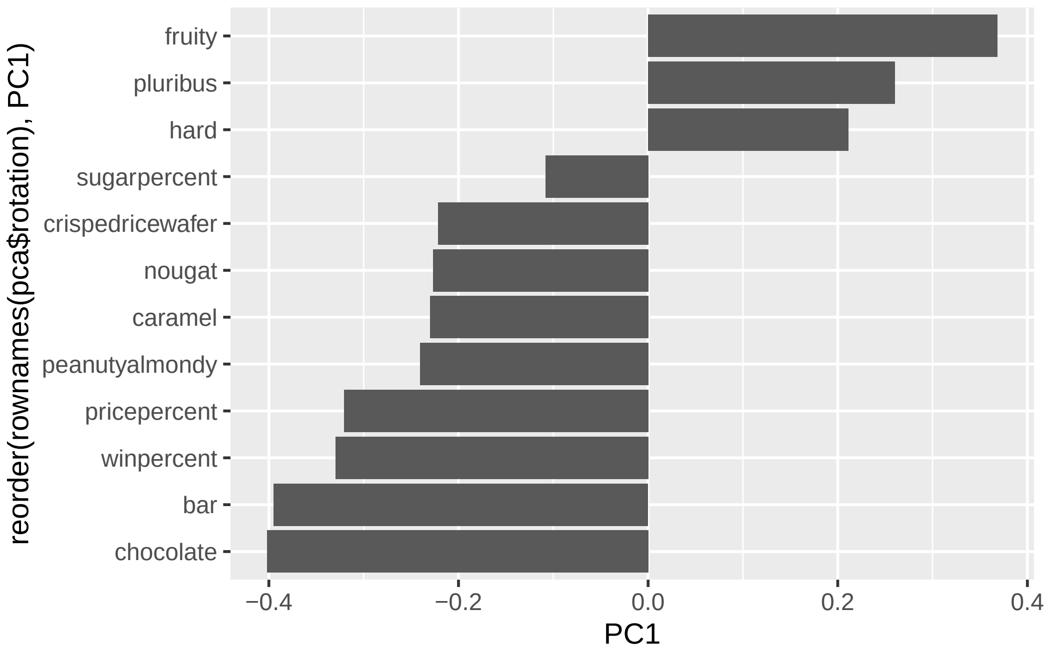



```r
library(ggrepel)

p + geom_text_repel(size=3.3, col=my_cols, max.overlaps = 7)  +
  theme(legend.position = "none") +
  labs(title="Halloween Candy PCA Space",
       subtitle="Colored by type: chocolate bar (dark brown), chocolate other (light brown),
       caption="Data from 538")
```




```
ggplot(as.data.frame(pca$rotation)) +
  aes(PC1, reorder(rownames(pca$rotation), PC1)) +
  geom_col()
```



Q24: Positive PC1 correlation

Fruity, Pluribus, and Hard are 3 variables picked up strongly in the positive direction

Summary

Q25: "Winning" candy characteristics

Based on the analyses done, people prefer chocolate candy bars that are affordable and potentially contain fillings like caramel, nougat, peanut butter, etc. Visualization presents patterns in preference, Correlation highlights variables of choice, and PCA is direct and reduces variation in answers.